

IT3071 – Machine Learning and Optimization Methods

Introduction to Artificial Intelligence and Machine Learning

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Introduction to Module

- This is a compulsory unit for the 3rd year students who follow BSc (Hons) in IT specializing in the Data Science program.
- The objective of this unit is to get an in-depth understanding of both the theory and practices in the field of Neural Networks and Optimization methods.

Introduction to Module

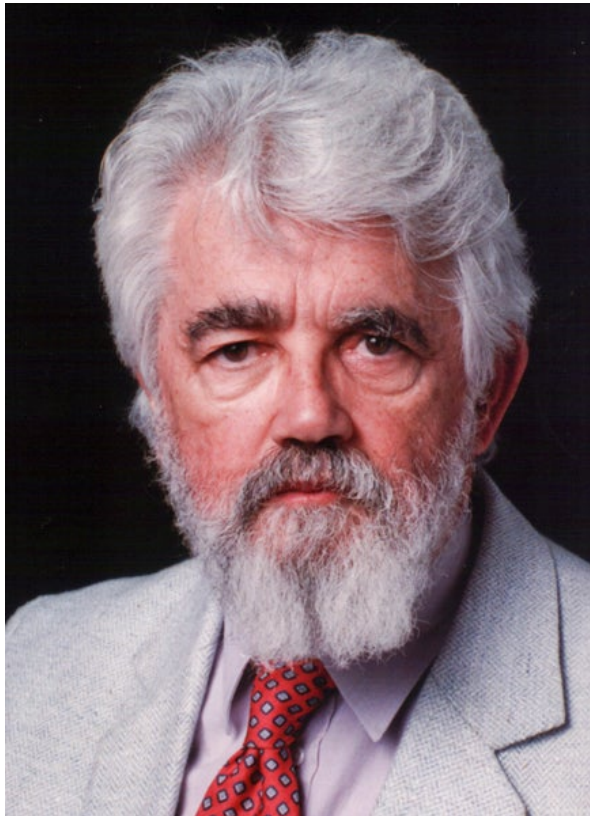
- In this unit, the students will learn:
 - Fundamentals of Neural Networks
 - Multilayer Neural Networks
 - Convolutional Neural Networks
 - Recurrent Networks
 - Reinforcement Learning
 - Support vector machines
 - Linear Programming
 - Genetic Algorithms

Assessments

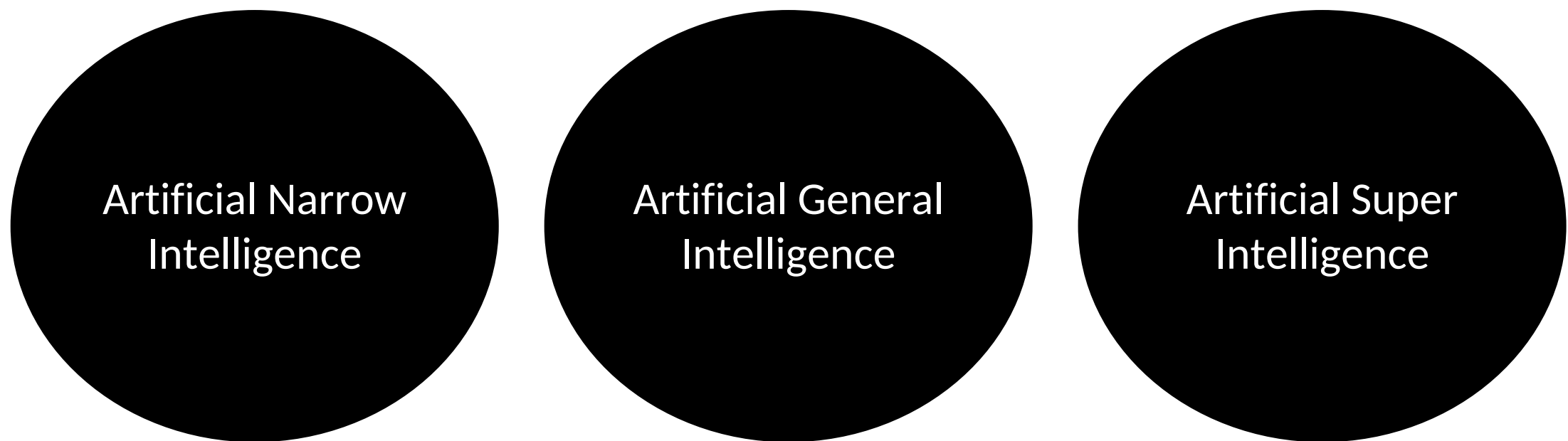
- Assignment 1 (25%) – In Class Practical Assignment (20%) + Lab Tasks (5%)
- Assignment 2 (25%) – Group Assignment (25%)
- Final Exam (50%)

What is Artificial Intelligence?

“The science and engineering of making intelligent machines, especially intelligent computer programs” – John McCarthy



Main Types of AI



Artificial Narrow
Intelligence

Artificial General
Intelligence

Artificial Super
Intelligence

Artificial Narrow Intelligence (ANI)

- Also referred to as weak AI or narrow AI, is the only type of artificial intelligence we have successfully realized to date.
- Goal-oriented, designed to perform singular tasks - i.e. facial recognition, speech recognition/voice assistants, driving a car, or searching the internet - and is very intelligent at completing the specific task it is programmed to do.
- They operate under a narrow set of constraints and limitations, which is why this type is commonly referred to as weak AI.

Artificial General Intelligence (AGI)

- Artificial general intelligence (AGI), also referred to as strong AI or deep AI, is the concept of a machine with general intelligence that mimics human intelligence and/or behaviours, with the ability to learn and apply its intelligence to solve any problem.
- AGI can think, understand, and act in a way that is indistinguishable from that of a human in any given situation.
- AI researchers and scientists have not yet achieved strong AI.
- AGI focuses on **Theory of Mind (TOM)** AI framework.

Artificial Super Intelligence (ASI)

- Artificial superintelligence (ASI) is a form of AI that is capable of surpassing human intelligence by manifesting cognitive skills and developing thinking skills of its own.
- Also known as super AI, artificial superintelligence is considered the most advanced, powerful, and intelligent type of AI that transcends the intelligence of some of the brightest minds, such as Albert Einstein.
- Machines with superintelligence are self-aware and can think of abstractions and interpretations that humans cannot.

Agents

- An agent refers to any entity (software or hardware) that perceives its environment through sensors and acts upon it using actuators, with the goal of achieving specific objectives.
- Two types of agents
 - Software Agents
 - These agents exist solely in a digital environment.
 - They can interact with other software, systems, or the internet and are often used to automate tasks, solve problems, or make decisions.
 - Ex – Siri, Netflix Recommendation System
 - Physical Agents
 - These agents are embodied in the physical world and can interact with it through mechanical means.
 - They perceive the environment through sensors (like cameras or infrared) and act upon it using actuators (like motors or arms).
 - Ex- Robots, Autonomous Drones, Self Driving Cars

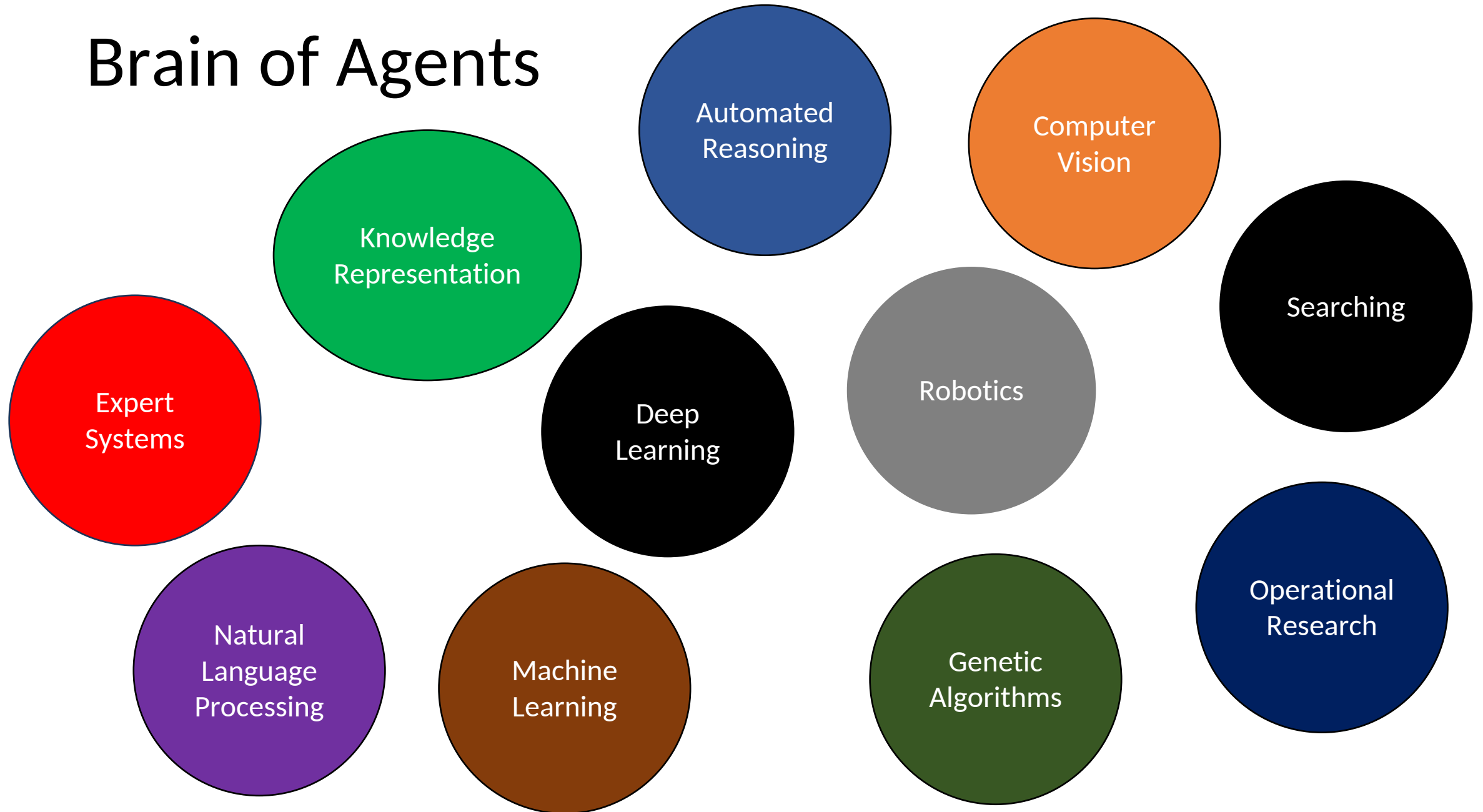
Single Agent Systems (SAS)

- A single-agent system is a system that consists of a single agent. An agent is an autonomous entity that can sense its environment, make decisions, and act to achieve its goals. Single-agent systems are often used to solve problems that are well-defined and have a single optimal solution.
- Examples of single-agent systems:
 - A chess-playing program
 - A robot that navigates a maze
 - A spam filter

Multi-Agent Systems (MAS)

- A multi-agent system (MAS) is a system that consists of multiple interacting agents. Agents in an MAS can cooperate or compete with each other in order to achieve their goals. Multi-agent systems are often used to solve problems that are complex and have multiple possible solutions.
- Examples of multi-agent systems:
 - A group of robots that collaborate to build a structure
 - A swarm of bees that collectively search for food
 - A team of players that cooperates to win a game

Brain of Agents



Hard Computing

- Hard computing refers to traditional, rule-based, deterministic computing where problems are solved using precise algorithms and logic.
- Key Features:
 - Requires complete, precise input
 - Gives exact solutions
 - Examples: A banking system verifying if account balance $\geq$ transaction amount using precise logic.

Soft Computing

- Soft computing is a branch of computing that uses approximate reasoning and probabilistic models to handle uncertainty, imprecision, and partial truth similar to how humans make decisions..
- Key Features:
 - Inspired by biological systems and human reasoning
 - Works with incomplete, noisy, or imprecise data
 - Produces acceptable (but not always perfect) solutions

Soft Computing

- Components
 - Fuzzy Logic
 - Artificial Neural Networks (ANNs)
 - Evolutionary Algorithms (like Genetic Algorithms)
 - Probabilistic Reasoning (like Bayesian networks)
- Example: Predicting whether a customer will buy a product based on past behavior, even if the data is incomplete or noisy.

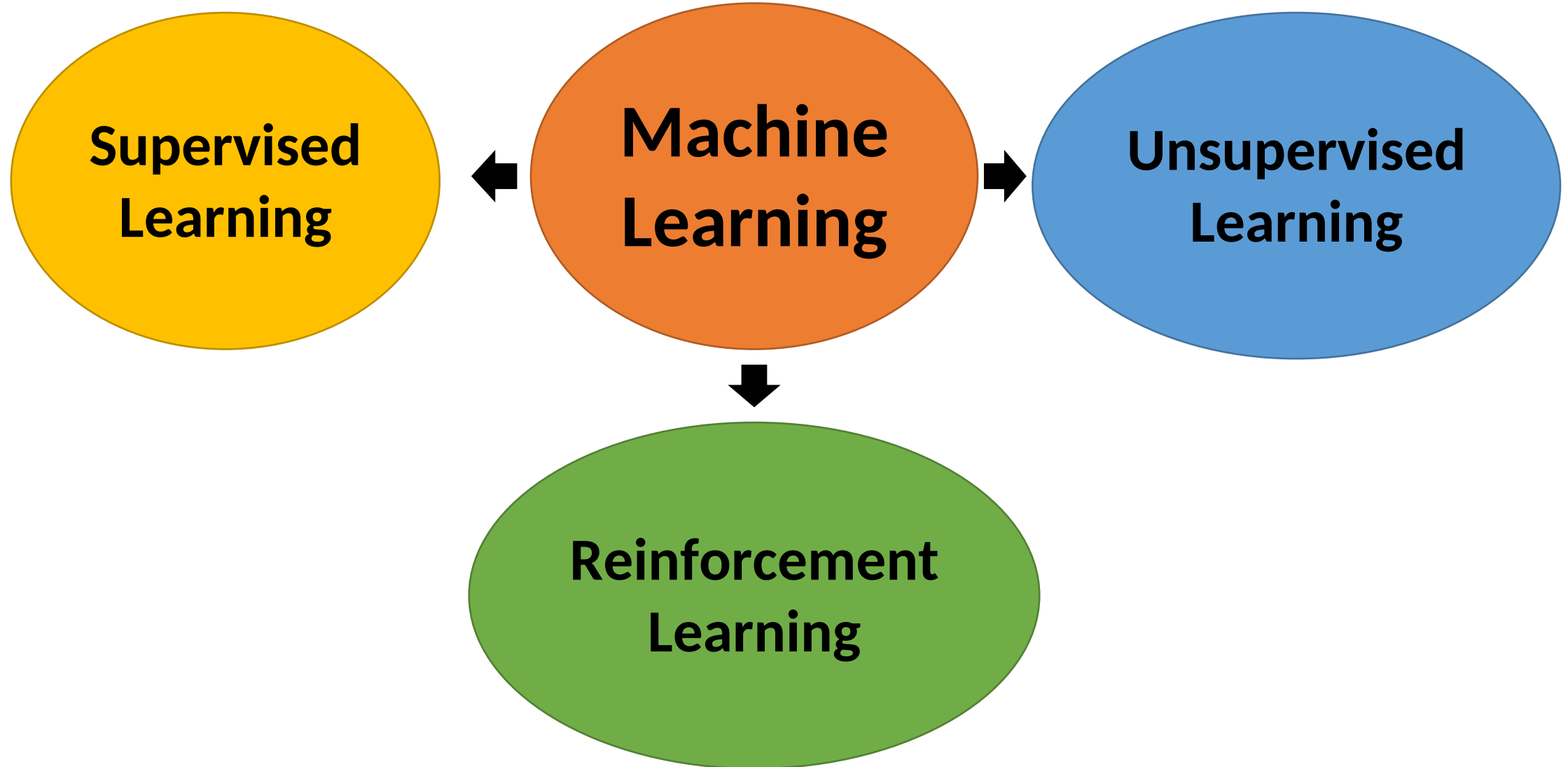
Hard Computing VS Soft Computing

Aspect	Hard Computing	Soft Computing → ML
Logic	Deterministic	Probabilistic / Approximate
Flexibility	Low	High
Handles Uncertainty	Poorly	Effectively
Rule Creation	Manual (rules coded)	Data-driven (learned from data)

Machine Learning

- ML algorithms learn patterns and relationships from data without being explicitly programmed.
- They improve their performance over time as they are exposed to more data.
- ML models represent knowledge implicitly in the form of statistical patterns learned from data. This knowledge is often stored in the parameters of the model.
- ML models (especially deep learning, probabilistic models, reinforcement learning, etc.) handle uncertainty, patterns, and generalization, key aspects of soft computing.

Machine Learning



Supervised Learning

- In Supervised Learning, teacher shows how to do
- Students learn by understanding the inputs & outputs
- The model will be trained with inputs and related outputs.
- After training it can make decisions based on predictions on unseen situations.



Underfitting and Overfitting

- **Overfitting** refers to a model that models the training data too well. This means that the noise or random fluctuations in the training data is picked up and learned as concepts by the model. The problem is that these concepts do not apply to new data and negatively impact the models ability to generalize.
- **Underfitting** refers to a model that can neither model the training data nor generalize to new data. An underfit machine learning model is not a suitable model and will be obvious as it will have poor performance on the training data.

Unsupervised Learning

- Unsupervised learning, is where you only have input data (X) and no corresponding output variables.
- Derive some structure or pattern from the “unlabeled data” by just looking at the relationship between the data themselves.

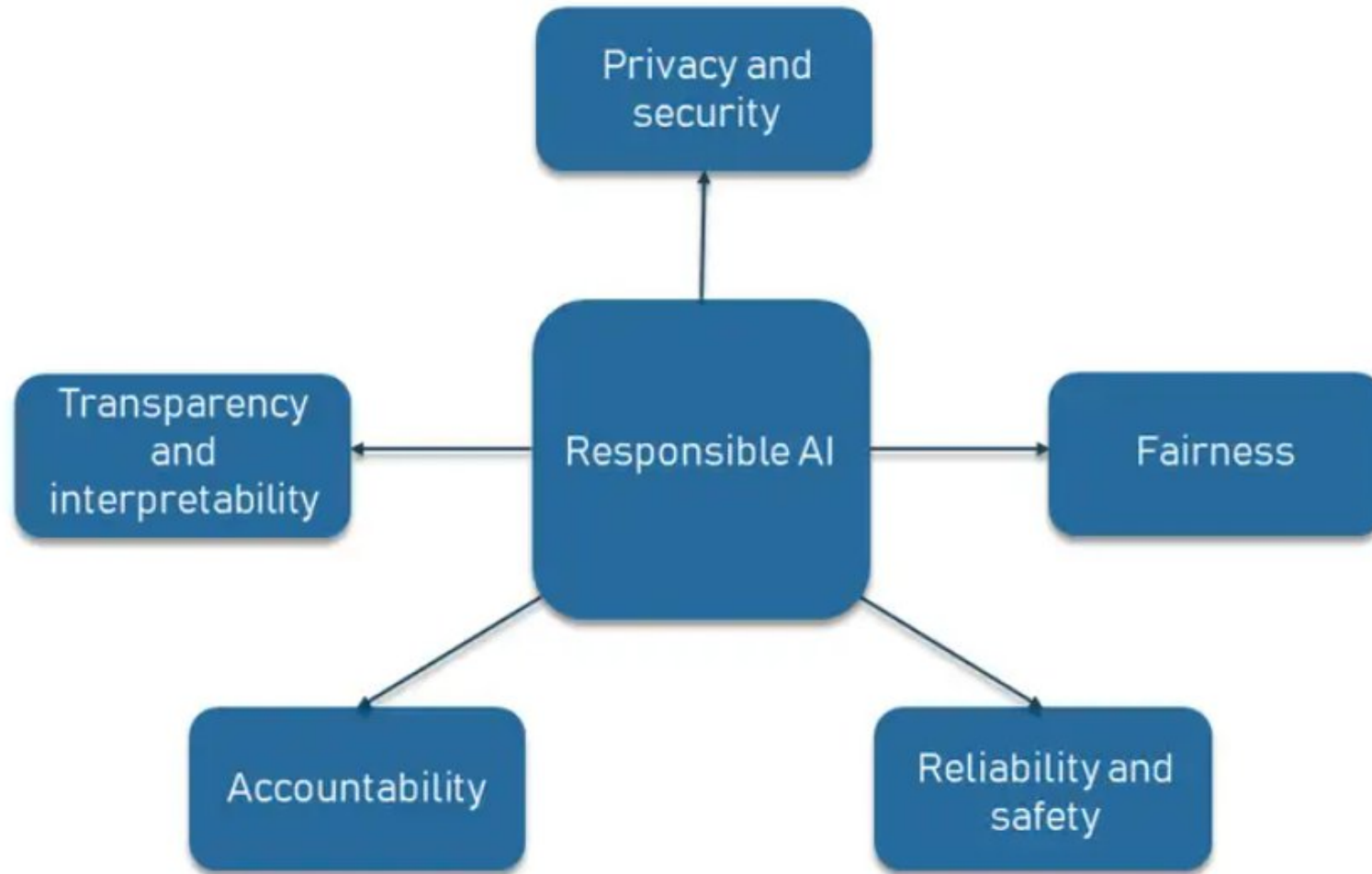


Reinforcement Learning

- It is about taking suitable action to maximize reward in a particular situation.
- It is employed by various software and machines to find the best possible behavior or path it should take in a specific situation.



Responsible AI



Privacy and Security

- Responsible AI ensures that user data is protected from unauthorized access and misuse.
- Privacy refers to protecting individual identity, while security refers to protecting data systems and models from attacks.
 - Data Anonymization: Remove personal identifiers from datasets.
 - Differential Privacy: Add noise to data to prevent tracing individual records.
 - Federated Learning: Train models on user devices without centralizing the raw data.
 - Encryption: Protect data during storage and transfer.
- Ex- Apple's Face ID uses on-device machine learning, ensuring your face data never leaves the device, maintaining user privacy.

Fairness

- Fairness in AI is about preventing discrimination or bias based on gender, race, religion, age, or any protected attribute.
- It promotes equal treatment and opportunity for all users.
 - Data Bias: Biased datasets from real-world inequalities.
 - Algorithmic Bias: Model amplifies or introduces bias.
 - Labeling Bias: Subjectivity in how data was labeled.
- Ex- A hiring AI trained on historical resumes might prefer male applicants. Amazon discontinued such a system after it penalized resumes with the word “women’s”.

Reliability and Safety

- AI systems should work consistently under varied conditions and fail safely if unexpected inputs are encountered.
- This is critical in real-world, high-risk settings.
 - Robust Training: Use adversarial training and diverse datasets.
 - Stress Testing: Simulate edge cases and real-world noise.
 - Fail-safe Mechanisms: Gracefully degrade the system when failure is imminent.
- Ex- Tesla's Autopilot must detect obstacles, road signs, and pedestrians reliably—even in poor lighting or weather. A failure could cause fatal accidents.

Accountability

- Clear mechanisms must be in place to trace actions and assign responsibility when something goes wrong.
- This is critical for legal, ethical, and operational safety.
 - Audit Trails: Keep logs of data usage and model behavior.
 - Version Control: Maintain traceable versions of training data and models.
 - Defined Roles: Know who is responsible—data provider, developer, deployer, or operator.
- Ex- If an AI model denies a loan unfairly, a human officer or the company should be accountable, not the machine. Laws like the EU AI Act stress this point.

Transparency and Interpretability

- Stakeholders (developers, users, regulators) should be able to understand how an AI model works, especially when decisions affect people's lives.
 - Explainable AI (XAI): Use methods like SHAP, LIME to visualize decision influences.
 - Model Cards: Document a model's purpose, performance, limitations.
 - Dataset Transparency: Document data sources, collection methods, and biases.
- Ex- A bank using AI for loan approvals must explain to rejected applicants why they were denied. Tools like SHAP can show that low income or credit score affected the decision most.